1. What is the main goal of approximation algorithms? A) To find the optimal solution B) To find a solution close to optimal C) To reduce the running time D) To increase the memory usage

2. Which of the following is a characteristic of approximation algorithms? A) They always return an optimal solution B) They always run in exponential time C) They always run in polynomial time D) They never return a solution close to optimal

3. What is the definition of an α-approximation algorithm? A) A polynomial-time algorithm that always returns a solution whose value ALG is within a factor α of the optimal value OPT B) A polynomial-time algorithm that always returns an optimal solution C) A polynomial-time algorithm that always returns a solution whose value ALG is greater than the optimal value OPT D) A polynomial-time algorithm that always returns a solution whose value ALG is less than the optimal value OPT

4. What is the approximation ratio of an algorithm? A) The smallest α such that the algorithm is an α-approximation algorithm B) The largest α such that the algorithm is an α-approximation algorithm C) The average value of α for which the algorithm is an α-approximation algorithm D) The maximum value of α for which the algorithm is not an α-approximation algorithm

5. What is the goal of the vertex cover problem? A) To find a minimum vertex cover of a graph B) To find a maximum independent set of a graph C) To find a minimum coloring of a graph D) To find a maximum flow of a graph

1. A
2. C
3. A
4. A
5. A